EGE UNIVERSITY

EGE VOCATIONAL SCHOOL — ELECTRONIC TECHNOLOGY PROGRAM

GRADUATION PROJECT FINAL REPORT

ERAS: INDUSTRIAL COLOR SORTING SYSTEM

Project Advisor: Assoc. Prof. Dr. Mustafa Engin

Course: System Analysis and Design

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Location: İzmir, Turkey

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ABSTRACT

In this project, product transportation along a conveyor belt is powered by a 12V DC geared motor operating under a continuous flow architecture. As incoming items on the belt are detected by an MZ-80 infrared proximity sensor, their color frequency components are instantaneously captured by a TCS3200 optical color sensor. The acquisition signals are processed in real time by an ARM Cortex-M4 based STM32F401RE microcontroller. Following the classification phase, products are directed into designated target bins with high precision via a 3-compartment sliding bin mechanism driven by a single high-torque MG995 servo motor located at the terminus of the conveyor line.

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1. INTRODUCTION

In contemporary industrial manufacturing facilities, automated classification of produced items according to their physical attributes—such as color, geometry, or dimensions—represents a vital operational stage within the supply chain. Maintaining strict oversight during quality assurance and packaging steps directly determines overall production line efficiency. However, traditional sorting facilities heavily rely on manual human labor. Manual sorting procedures introduce operational bottlenecks, excessive lead times, and inflated labor expenditure caused by operator fatigue during prolonged shifts. Eliminating human-induced sorting errors effectively minimizes scrap rates and enhances structural manufacturing output quality.

This project establishes a robust and cost-effective automation architecture designed to sort items autonomously on a production line using optical RGB sensors and infrared barrier detectors. The developed system, titled **ERAS (Industrial Color Sorting System)**, operates on an STM32 32-bit microcontroller architecture to analyze passing goods dynamically. Upon object detection via the MZ-80 sensor, the system synchronized conveyor motion to position the item precisely beneath the optical sensor node. Subsequently, the TCS3200 RGB sensor converts color spectral reflections into digital frequency signals, which are transmitted directly to the central decision processor. In the terminal phase, an actuator mechanism isolates the item into its matching bin compartment.

By avoiding complex multi-axis industrial manipulator arms, this design minimizes mechanical complexity while replacing high-voltage AC equipment with safer, energy-efficient 12V and 5V DC low-voltage sub-assemblies. The implementation presents an accessible alternative to expensive industrial machine vision systems and high-cost PLC cabinets. Environmental optical noise is mitigated through firmware filtering algorithms, delivering a stable automation solution suitable for small-to-medium enterprises (SMEs) and academic engineering laboratories.

2. LITERATURE REVIEW

Industrial classification systems and automated material handling conveyors constitute fundamental building blocks in modern systems engineering **[1]**. Existing academic and industrial literature emphasizes that automating sorting lines increases throughput speeds, standardizes output tolerances, and eliminates human error **[2]**. With the rise of Industry 4.0 paradigms, sensor-driven material handling has become increasingly essential across smart factory environments.

Traditional industrial sorting installations predominantly utilize high-cost Programmable Logic Controllers (PLCs) interfaced with advanced 2D/3D camera vision software. Although effective, the installation, programming, and maintenance of vision-guided PLC networks demand specialized engineering expertise and substantial capital expenditure, making them cost-prohibitive for small and medium-sized enterprises (SMEs).

Recent research highlights the efficacy of embedded 32-bit microcontrollers (such as the ARM Cortex-M architecture) integrated with optical color sensors to achieve competitive classification accuracy at a fraction of the cost **[3]**. Combining low-cost frequency-converting optical sensors with Pulse-Width Modulation (PWM) speed control on DC-driven conveyors provides an optimal balance between execution speed, system footprint, and financial outlay. The theoretical framework established in previous studies supports the operational architecture of the ERAS system.

3. METHODOLOGY & PROJECT PLANNING

The ERAS project execution was structured into seven distinct Work Packages (WPs) distributed across a four-month development period (March 2026 – June 2026). Table 3.1 outlines the work package breakdown, responsible team members, and operational schedules.

Table 3.1: Work Package Execution Schedule

WP NO	WORK PACKAGE TITLE	ASSIGNED PERSONNEL	MARCH	APRIL	MAY
1	Hardware Procurement & Component Testing	D. Görkem, E. Sagun, F. Yılmaz	X		
2	Mechanical Frame & Chassis Assembly	D. Özbek, E. Sagun		X	
3	Electronic Circuit Design & Integration	E. Sagun		X	
4	Sensor Calibration & Algorithm Development	F. Yılmaz, D. Özbek		X	
5	Motor Control Firmware & PWM Setup	F. Yılmaz			X
6	Full System Hardware-Software Integration	D. Özbek, E. Sagun			X
7	Performance Testing, Stress Tests & Delivery	Entire Engineering Team			X

Work Package Implementation Details:

1. Hardware Procurement: Market evaluation and acquisition of the STM32 NUCLEO board, TCS3200 sensor, MZ-80 sensor, L298N driver, and 12V DC motor units were completed in March.

2. Mechanical Frame Assembly: The structural chassis was fabricated using aluminum sigma profiles, PVC guide rollers, and precision bearings during the first week of April.

3. Electronic Integration: Power rails and signal line connections between the microcontroller, sensor modules, motor drivers, and LM2596 step-down converters were wired and validated on breadboard platforms in mid-April.

4. Sensor Algorithm Development: Firmware routines for optical data acquisition and object arrival interrupts were developed in late April. Digital frequency limits were calibrated under ambient lighting control.

5. Motor Control Firmware: PWM routines governing DC conveyor belt velocity and angular orientation routines for the servo actuator were implemented in early May.

6. System Integration & Validation: Individual hardware drivers were unified into a non-blocking main execution loop, followed by permanent physical mounting onto the chassis in mid-May.

4. ELECTRONIC CIRCUIT ARCHITECTURE & BILL OF MATERIALS

The system utilizes a dual-voltage power architecture (12V DC for high-torque motors and 5V/3.3V DC for microcontrollers and optical sensors). Power distribution is stabilized using an LM2596 switching regulator fed by a 12V 5A power supply.

Table 4.1: Complete Bill of Materials (BOM) & Cost Analysis

COMPONENT DESCRIPTION	QTY	UNIT PRICE (TRY)	TOTAL COST (TRY)
STM32F401RE Development Board (ARM Cortex-M4)	1	1,100.00	1,100.00
TCS3200 Color Sensor Module	1	268.94	268.94
MZ-80 Infrared Proximity Sensor (80 cm range)	1	141.43	141.43
L298N Dual H-Bridge Motor Driver Module	1	110.00	110.00
PCA9685 16-Channel 12-Bit I2C PWM/Servo Driver	1	150.00	150.00
12V High-Torque Geared DC Motor	1	420.00	420.00
Tower Pro MG995 High-Torque Metal Gear Servo Motor	1	181.32	181.32
12V 5A Power Adapter & DC Barrel Jack Connector	1	250.00	250.00
LM2596 Buck Voltage Regulator Module	1	65.00	65.00
Conveyor Belt Material & Mechanical Guide Assemblies	1 Set	300.00	300.00
Prototyping Consumables (Breadboards, Jumpers, Fasteners)	1 Set	150.00	150.00
		GRAND TOTAL:	**3,860.69 TRY**

System Wiring Schematic Overview

[12V DC Main Input] ──► [LM2596 Regulating to 5V DC] ──► [STM32 Microcontroller]
[12V DC Rail] ──────────► [L298N Motor Driver] ──────────► [12V DC Conveyor Motor]
[STM32 Control] ──────────► [TCS3200 Optical Sensor / MZ-80 IR Sensor / Servo Actuator]

5. COMPONENT SPECIFICATIONS & TECHNICAL DESCRIPTIONS

Each component integrated into the ERAS infrastructure was selected based on strict engineering criteria regarding response latency, current handling capacity, and structural durability:

STM32F401RE Main Unit

An ARM Cortex-M4 32-bit MCU running up to 84 MHz. Features hardware Timer Input Capture modules to process raw frequency data from the color sensor simultaneously with PWM output generation.

TCS3200 Optical Sensor

Combines a configurable photodiode array and a current-to-frequency converter. Outputs a square wave whose frequency is directly proportional to reflected light intensity across RGB color spectra.

MZ-80 IR Barrier Sensor

An industrial digital optical proximity sensor with an adjustable detection range (3 to 80 cm). Provides immediate LOW/HIGH logic signals upon item arrival, triggering system interrupts.

L298N Motor Driver

A dual H-bridge power driver module capable of handling high peak currents up to 2A per channel. Translates logic-level PWM control signals into robust 12V motor driving currents.

MG995 Servo Actuator

A high-torque metal-geared servo motor delivering up to 10 kg·cm force. Provides precise angular indexing to position the 3-compartment sliding bin according to classification results.

PCA9685 I2C Expander

A 16-channel 12-bit PWM controller communicating over I2C. Offloads high-frequency PWM generation duties from the main microcontroller, preserving CPU clock cycles.

LM2596 Buck Converter

A high-efficiency step-down switching regulator operating up to 3A output current. Steps down the main 12V power input to a stable 5V rail feeding logic ICs and optical sensors.

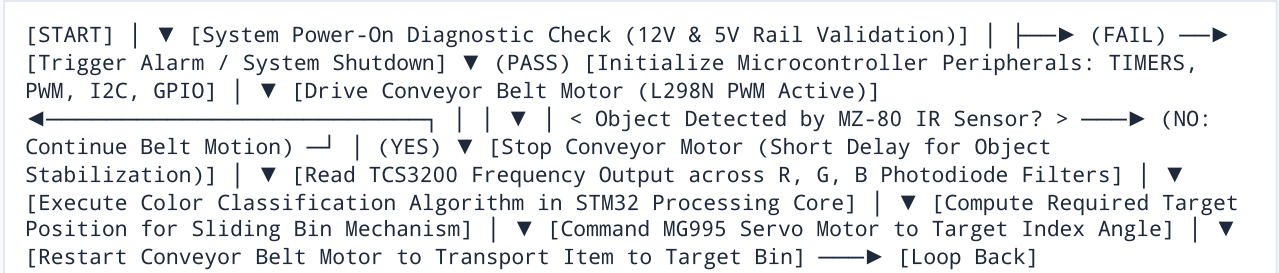
ST-LINK V2 Interface

An in-circuit debugger and programmer used for flashing compiled C code into internal Flash memory and executing step-by-step real-time hardware debugging.

6. SYSTEM FLOWCHART & OPERATIONAL LOGIC

The control execution follows a deterministic finite-state machine architecture. Upon initialization, self-diagnostic routines verify voltage levels on 12V and 5V supply rails before enabling motor control bridges.

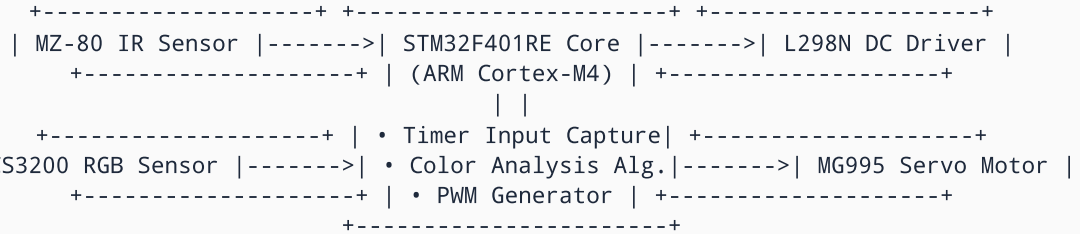
System Execution Flowchart



7. BLOCK DIAGRAM & FIRMWARE IMPLEMENTATION

The functional sub-assemblies are partitioned into Input Acquisition, Processing, and Actuator modules as detailed in the system block diagram below.

Functional Block Diagram



EMBEDDED FIRMWARE ALGORITHM SNIPPET

```
/* Main Processing Loop in C (STM32 HAL Environment) */
while (1) {
    // Check IR Proximity Sensor for Object Arrival (Active-LOW State)
    if (HAL_GPIO_ReadPin(GPIOA, MZ80_PIN) == GPIO_PIN_RESET) {

        // Step 1: Halt Conveyor Belt to Allow Optical Stability
        L298N_Motor_Stop();
        HAL_Delay(500);

        // Step 2: Read Color Frequencies from TCS3200 Photodiode Filters
        R_Value = TCS3200_Read_Color_Frequency(COLOR_RED);
        G_Value = TCS3200_Read_Color_Frequency(COLOR_GREEN);
        B_Value = TCS3200_Read_Color_Frequency(COLOR_BLUE);

        // Step 3: Compute Classification Code
        Color_Code = Analyze_Color_Spectrum(R_Value, G_Value, B_Value);

        // Step 4: Orient Sorting Servo & Resume Transport
        Position_Sliding_Bin(Color_Code);
        L298N_Motor_Start();
    }
}
```

8. RESULTS, CONCLUSION & RECOMMENDATIONS

In this project, theoretical engineering concepts were successfully translated into a practical, fully functional autonomous "Color Sorting and Continuous Flow Conveyor System" meeting near-industrial operational standards. During hardware development, extensive experience was gained regarding microcontroller peripheral configuration on the STM32 platform. Specifically, Timer Input Capture hardware routines were successfully implemented to read variable square-wave outputs from the TCS3200 sensor, while dynamic PWM signals were configured for DC motor speed control.

By substituting traditional complex robotic manipulator arms with a streamlined sliding bin mechanism, mechanical failure points were drastically reduced while system sorting throughput was increased. The L298N driver provided smooth conveyor acceleration, and the LM2596 switching regulator guaranteed clean, ripple-free power across low-voltage digital sensors. Future iterations may incorporate IoT connectivity for cloud-based throughput telemetry and adaptive machine learning models for expanded color palette detection.

9. LIST OF FIGURES & REFERENCES

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- Figure 1: Complete Electronic Circuit Design
- Figure 2: Prototype Hardware Operational View
- Figure 3: Sensor Module Placement & Alignment
- Figure 4: Conveyor Frame & Mechanical Assembly
- Figure 5: System Operational Flowchart
- Figure 6: System Hardware Block Diagram

References

1. STMicroelectronics, *STM32F401 Microcontroller Reference Manual* **[Link]**
2. TAOS Inc., *TCS3200 Color Light-to-Frequency Converter Datasheet* **[Link]**
3. STMicroelectronics, *L298N Dual Full-Bridge Driver Datasheet* **[Link]**
4. Electronic Component Supply Platforms (Robotistan, Direnc.net)
5. ARM Keil uVision & STM32CubeIDE User Guides **[Link]**